Mechanistic Lens on Mode Connectivity

Ekdeep Singh Lubana[1,2,4], Eric J. Bigelow[2], Robert P. Dick[1], David Krueger[3], Hidenori Tanaka[2,4]
[1]University of Michigan, [2]CBS Harvard University, [3]University of Cambridge, [4]NTT Research

Connectivity of Minimizers from a Given Dataset

- Mode Connectivity: Minimizers of modern NNs retrieved using SGD on a dataset can be connected via simple paths of zero loss.

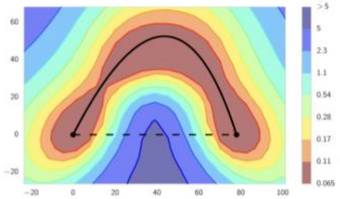
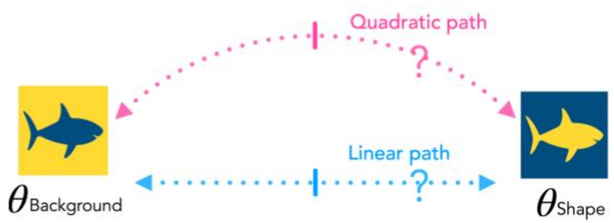


Figure: Train loss of ResNet-164 on CIFAR-100. (Garipov et al.)

Linear: $t\theta_1 + (1-t)\theta_2$;

Quadratic: $t^2\theta_1 + 2t(1-t)\theta_m + (1-t)^2\theta_2$.

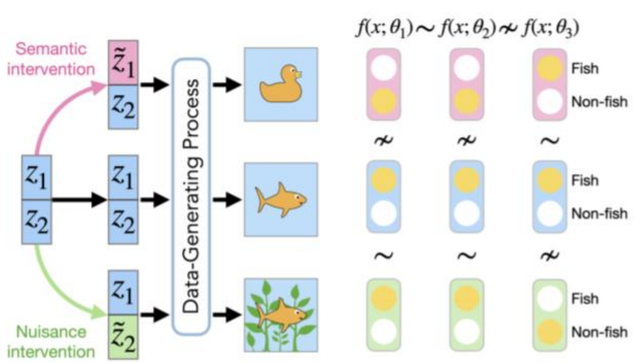
Motivating Problem: Are Minimizers Retrieved from Different Datasets of a Task Connected?



- Q1: Are mechanistically dissimilar models, which rely on different input attributes to make predictions, connected?
- Q2: Can we exploit this connectivity?

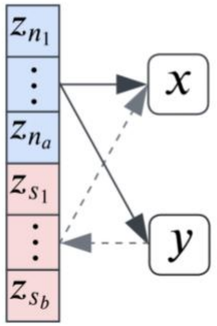
Towards a Definition of Mechanistic Similarity

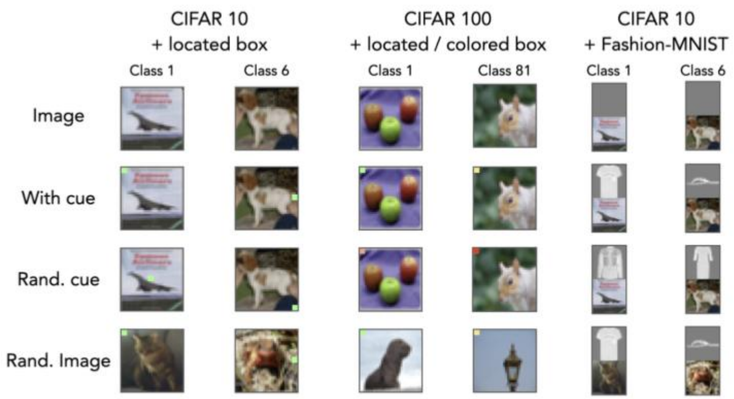
- We use shared invariances to interventions on the data-generating process to define a notion of ***Mechanistic Similarity*** between two models.



Synthetically Inducing Mechanistically Dissimilar Models

- A data-generating process that synthetically embeds controllable, partially predictive cues in the data.
- Evaluation: Counterfactuals that break predictivity of label by altering synthetic versus natural attributes.
- Goal: Analyze connectivity of models that rely vs. ignore spurious attributes.





Result 1: Are Mechanistically Dissimilar Minimizers Connected? Yes!

Proposition. ***(Mode Connectivity under Mechanistic Dissimilarity.)*** *Assume θ_1, θ_2 are minimizers of the loss on a dataset $\mathcal{D}$ and induce mechanistically dissimilar models. Given sufficient over-parameterization, there exists a continuous path along which the minimizers are mode connected.*

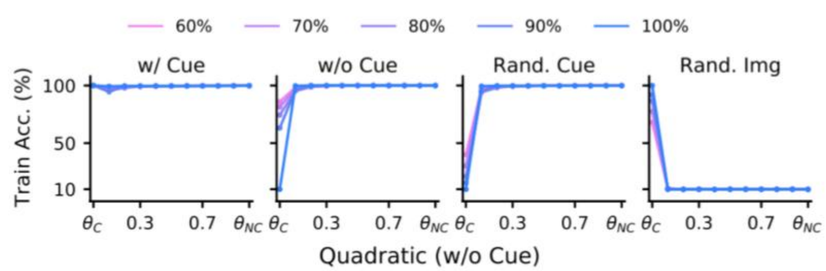


Figure: ResNet-18 models trained on located cue CIFAR-10 with different % cue data (C) show connectivity via quadratic paths to modes identified using no cue (NC) data, but are mechanistically dissimilar.

Result 2: Is Mechanistic Similarity and Simplicity of Connectivity Paths Related? Yes!

Conjecture. ***(Lack of Linear Connectivity implies Mechanistic Dissimilarity.)*** *If two minimizers θ_1 and θ_2 of the loss $\mathcal{L}(f(\mathcal{D};\theta))$ on a dataset $\mathcal{D}$ cannot be linear mode connected (up to architectural symmetries), their induced models $f(.;\theta_1), f(.;\theta_2)$ must be mechanistically dissimilar.*

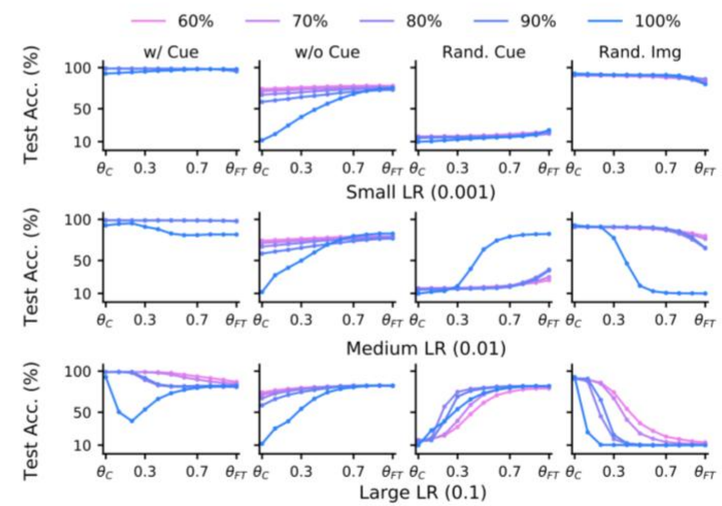
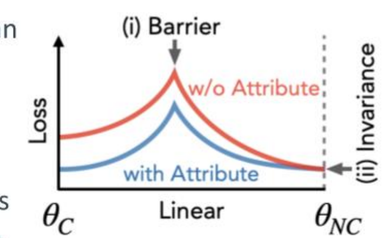


Figure: ResNet-18 models trained on located cue CIFAR-10 (C) and fine-tuned on data w/o cue (FT) show linear connectivity under small–medium learning rates, yielding mechanistically similar modes.

Result 3: Altering Mechanisms via Lack of Linear Connectivity

- Assume a model learns to rely on an undesirable input attribute: Can we "alter" its mechanisms to use another, desirable attribute?
- Connectivity-Based Fine-Tuning (CBFT): Find a minimizer that lacks linear connectivity but is invariant to the undesirable attribute.



	60% Cue data				70% Cue data				80% Cue data				90% Cue data			
C-10	NC↑	C~	RC~	RI↓	NC↑	C~	RC~	RI↓	NC↑	C~	RC~	RI↓	NC↑	C~	RC~	RI↓
FT_M	**75.7**	98.4	23.6	83.4	**75.8**	98.6	27.7	78.6	71.3	97.7	37.6	63.6	67.2	95.4	49.6	46.6
FT_S	75.8	98.7	17.5	90.1	74.9	98.8	16.3	91.1	69.9	98.4	15.7	90.9	64.7	97.9	15.3	90.7
LLR	71.6	95.1	36.3	57.1	70.9	95.8	29.9	65.8	65.1	81.8	27.0	53.2	59.3	70.7	24.6	40.7
LPFT	70.6	88.1	21.0	70.7	69.6	87.3	18.7	72.5	64.4	63.8	18.8	48.0	59.7	56.6	19.8	37.8
CBFT	74.1	**71.5**	**73.4**	**8.75**	73.2	**69.2**	**72.3**	**8.60**	70.0	**70.0**	**69.5**	**9.68**	**67.9**	**72.5**	**68.1**	**13.1**